Jaire Bartlett

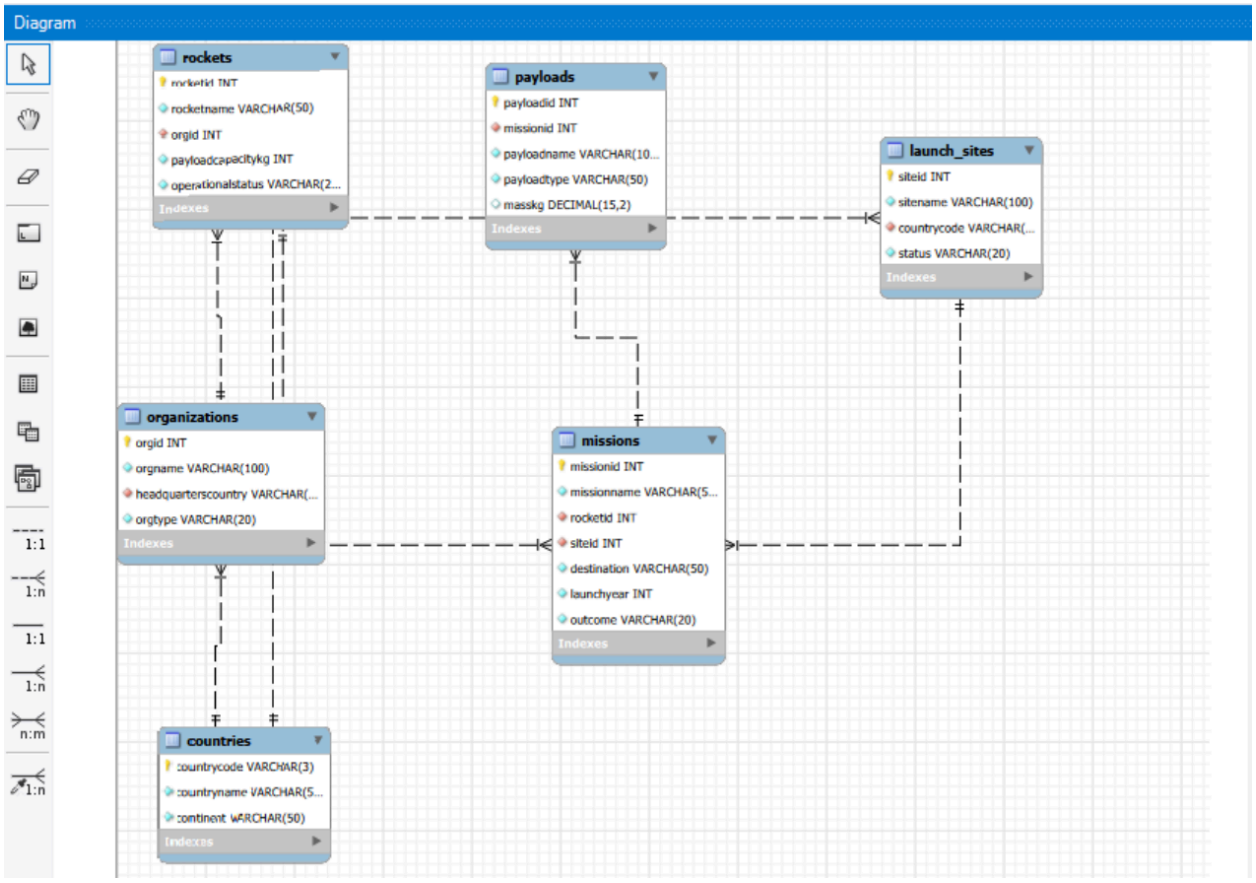
CSC 351

24 October 2026

Homework Assignment 4

Building a Global Space Missions and Launch Operations Relational Model in MySQL

Full Diagram:



SQL Queries & Screenshots

SQL Queries:

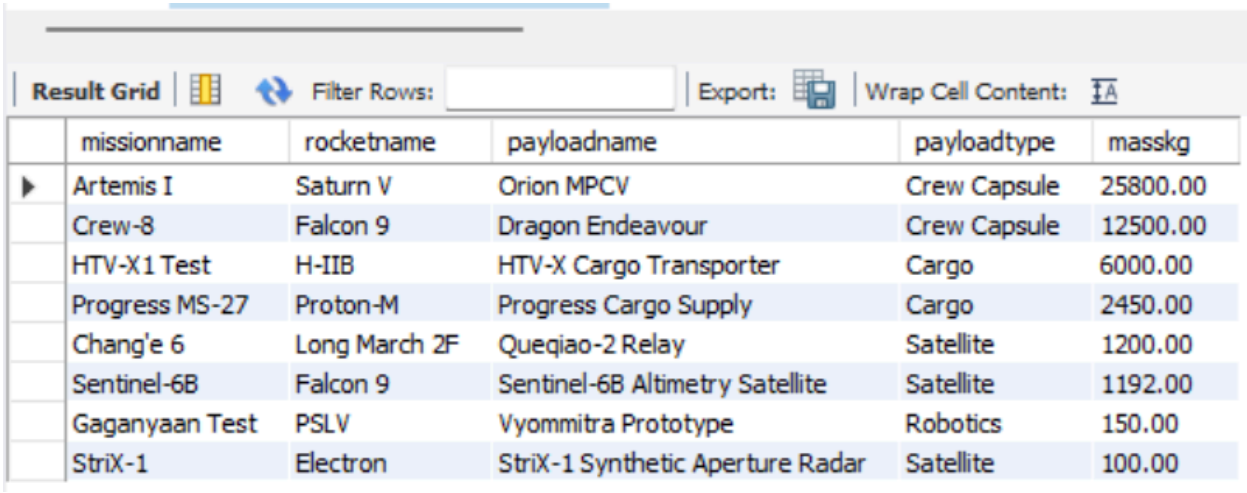
1. Retrieve the mission name, rocket name, organization name, launch site name, destination, launch year, and outcome for all space missions without using table aliases. Sort the results in descending order by launch year and ascending order by mission name.
    a.
```
select
    missions.missionname,
    rockets.rocketname,
    organizations.orgname,
    launch_sites.sitename,
    missions.destination,
    missions.launchyear,
    missions.outcome
from
    missions
join
    rockets on missions.rocketid = rockets.rocketid
join
    organizations on rockets.orgid = organizations.orgid
join
    launch_sites on missions.siteid = launch_sites.siteid
order by
    missions.launchyear desc,
    missions.missionname asc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

missionname	rocketname	orgname	sitename	destination	launchyear	outcome
Artemis I	Saturn V	National Aeronautics and Space Administration	Kennedy Space Center	Moon	2024	Success
Chang'e 6	Long March 2F	China National Space Administration	Jiuquan Satellite Launch Center	Moon	2024	Success
Crew-8	Falcon 9	Space Exploration Technologies Corp	Kennedy Space Center	ISS	2024	Success
Gaganyaan Test	PSLV	Indian Space Research Organisation	Satish Dhawan Space Centre	LEO	2024	Success
HTV-X1 Test	H-IIB	Japan Aerospace Exploration Agency	Tanegashima Space Center	ISS	2024	Success
Progress MS-27	Proton-M	Roscosmos State Corporation	Baikonur Cosmodrome	ISS	2024	Success
Sentinel-6B	Falcon 9	Space Exploration Technologies Corp	Vandenberg Space Force Base	LEO	2024	Success
StriX-1	Electron	Rocket Lab	Mahia Launch Complex 1	LEO	2024	Success

2. Retrieve the mission name, rocket name, payload name, payload type, and payload mass (in kg) for all payloads launched in 2024 without using table aliases. Sort the results in descending order by payload mass.
    a.
```
select
    missions.missionname,
    rockets.rocketname,
    payloads.payloadname,
    payloads.payloadtype,
    payloads.masskg
```

```
from
    payloads
join
    missions on payloads.missionid = missions.missionid
join
    rockets on missions.rocketid = rockets.rocketid
where
    missions.launchyear = 2024
order by
    payloads.masskg desc;
```

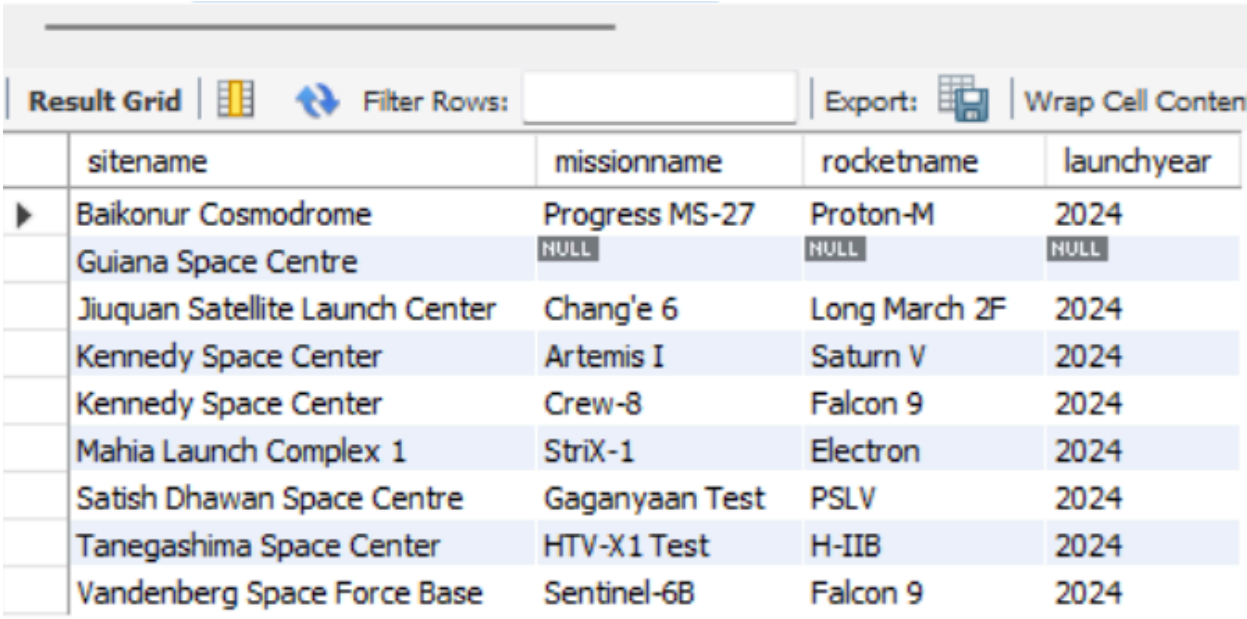


Result Grid | Filter Rows: | Export: | Wrap Cell Content:

missionname	rocketname	payloadname	payloadtype	masskg
Artemis I	Saturn V	Orion MPCV	Crew Capsule	25800.00
Crew-8	Falcon 9	Dragon Endeavour	Crew Capsule	12500.00
HTV-X1 Test	H-IIB	HTV-X Cargo Transporter	Cargo	6000.00
Progress MS-27	Proton-M	Progress Cargo Supply	Cargo	2450.00
Chang'e 6	Long March 2F	Queqiao-2 Relay	Satellite	1200.00
Sentinel-6B	Falcon 9	Sentinel-6B Altimetry Satellite	Satellite	1192.00
Gaganyaan Test	PSLV	Vyommitra Prototype	Robotics	150.00
StriX-1	Electron	StriX-1 Synthetic Aperture Radar	Satellite	100.00

3. Retrieve all launch sites along with their matching mission names, rockets, and launch years for missions that took place in 2024 without using table aliases, ensuring that launch sites with no recorded 2024 missions still appear in the results with null values for the mission fields. Sort the results in alphabetical order by launch site name.
    a.
```
select
    launch_sites.sitename,
    missions.missionname,
    rockets.rocketname,
    missions.launchyear
from
    launch_sites
left join
    missions on launch_sites.siteid = missions.siteid and missions.launchyear =
2024
left join
    rockets on missions.rocketid = rockets.rocketid
```

```
order by
    launch_sites.sitename asc;
```

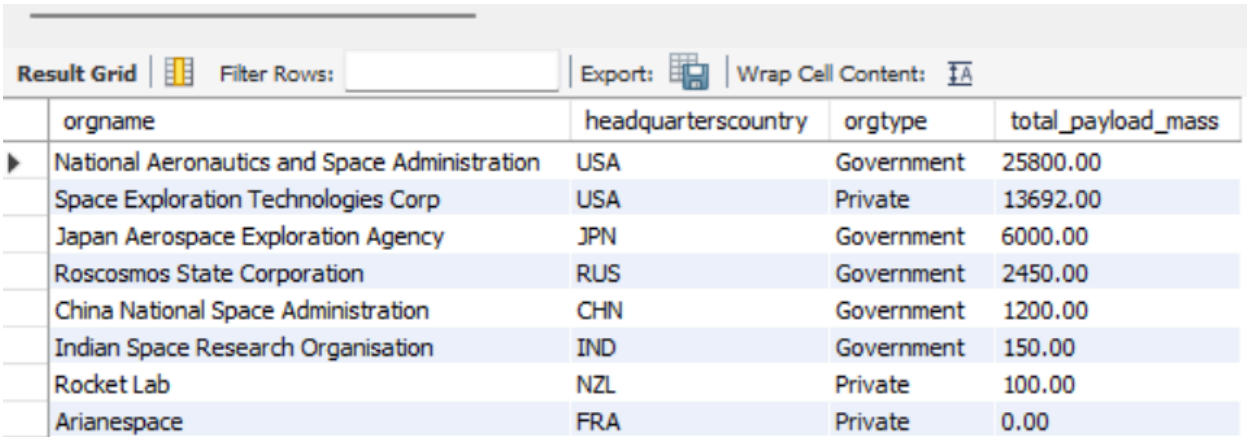


Result Grid | Filter Rows: | Export: | Wrap Cell Conten

sitename	missionname	rocketname	launchyear
Baikonur Cosmodrome	Progress MS-27	Proton-M	2024
Guiana Space Centre	NULL	NULL	NULL
Jiuquan Satellite Launch Center	Chang'e 6	Long March 2F	2024
Kennedy Space Center	Artemis I	Saturn V	2024
Kennedy Space Center	Crew-8	Falcon 9	2024
Mahia Launch Complex 1	StriX-1	Electron	2024
Satish Dhawan Space Centre	Gaganyaan Test	PSLV	2024
Tanegashima Space Center	HTV-X1 Test	H-IIB	2024
Vandenberg Space Force Base	Sentinel-6B	Falcon 9	2024

4. Write a SQL query using a Common Table Expression (CTE) to calculate the total payload mass (in kg) deployed by each organization across all missions without using table aliases. Retrieve the organization name, headquarters country, organization type, and total payload mass, and sort the results in descending order by total payload mass.
    a.
```
with org_payload_totals as (
    select
        rockets.orgid,
        sum(payloads.masskg) as total_mass
    from
        payloads
    join
        missions on payloads.missionid = missions.missionid
    join
        rockets on missions.rocketid = rockets.rocketid
    group by
        rockets.orgid
)
select
    organizations.orgname,
    organizations.headquarterscountry,
    organizations.orgtype,
```

```
    coalesce(org_payload_totals.total_mass, 0) as total_payload_mass
from
    organizations
left join
    org_payload_totals on organizations.orgid = org_payload_totals.orgid
order by
    total_payload_mass desc;
```

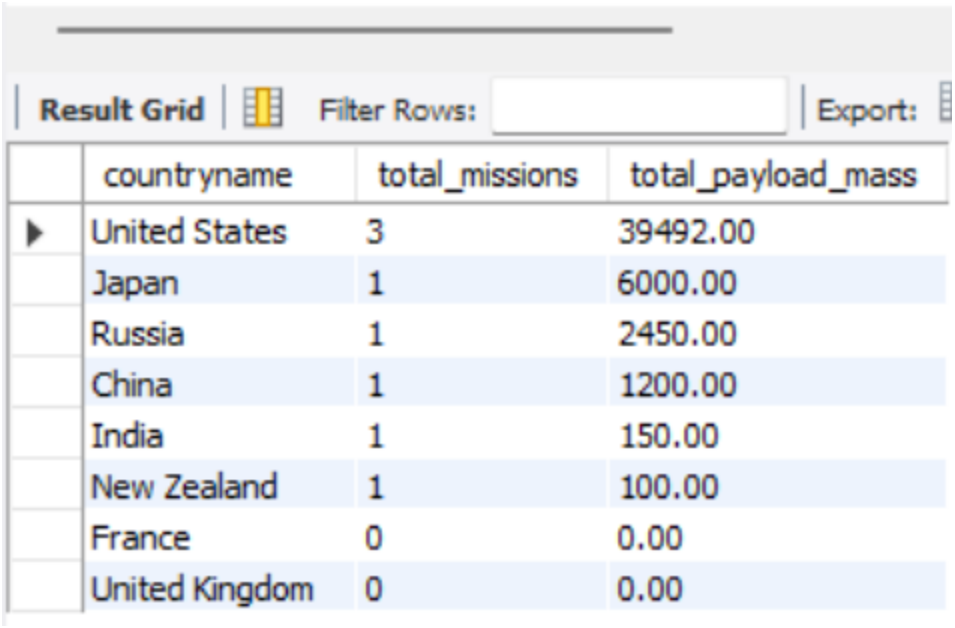


Result Grid | Filter Rows: | Export: | Wrap Cell Content:

orgname	headquarterscountry	orgtype	total_payload_mass
National Aeronautics and Space Administration	USA	Government	25800.00
Space Exploration Technologies Corp	USA	Private	13692.00
Japan Aerospace Exploration Agency	JPN	Government	6000.00
Roscosmos State Corporation	RUS	Government	2450.00
China National Space Administration	CHN	Government	1200.00
Indian Space Research Organisation	IND	Government	150.00
Rocket Lab	NZL	Private	100.00
Arianespace	FRA	Private	0.00

5. Write a SQL query using two separate Common Table Expressions (CTEs)—one to calculate total missions by country code and another to calculate total payload mass by country code without using table aliases. Then, combine these CTEs with the countries table to display the country name, total number of missions, and total payload mass (in kg). Sort the results in descending order by total payload mass.
    a.
```
with country_mission_counts as (
    select
        launch_sites.countrycode,
        count(missions.missionid) as total_missions
    from
        missions
    join
        launch_sites on missions.siteid = launch_sites.siteid
    group by
        launch_sites.countrycode
),
```

```
        country_payload_mass as (
            select
                launch_sites.countrycode,
                sum(payloads.masskg) as total_mass
            from
                payloads
            join
                missions on payloads.missionid = missions.missionid
            join
                launch_sites on missions.siteid = launch_sites.siteid
            group by
                launch_sites.countrycode
        )
        select
            countries.countryname,
            coalesce(country_mission_counts.total_missions, 0) as total_missions,
            coalesce(country_payload_mass.total_mass, 0) as total_payload_mass
        from
            countries
        left join
            country_mission_counts on countries.countrycode =
country_mission_counts.countrycode
        left join
            country_payload_mass on countries.countrycode =
country_payload_mass.countrycode
        order by
            total_payload_mass desc;
```


Result Grid | Filter Rows: | Export:

countryname	total_missions	total_payload_mass
United States	3	39492.00
Japan	1	6000.00
Russia	1	2450.00
China	1	1200.00
India	1	150.00
New Zealand	1	100.00
France	0	0.00
United Kingdom	0	0.00